The Basin Rule: One Surface-Density Condition for Where Cold Dark Energy Condenses, with Sealed Lensing Predictions and Audited Failures

Working note, v1.2 (2026-10-29; v1.1 and v1.0 2026-10-28/29). AI-assisted research programme; not peer reviewed.
$\kappa = \frac{1}{2}$ is fitted; the theory is not closed.

Correction in v1.1. v1.0's central claim, that "the threshold mass is not a separate number: it follows from L", is **withdrawn**. A parallel audit (CSM1) found that the one-number variant (CL3) evaluated the switch with a constant concentration coefficient, while the halo cores used the mass-dependent native slope. Made consistent, the switch drops to $\log T \approx 11.22$ and isolated-lens lensing fails by $\Delta\chi^2 = +415.281$ (allowed +9). The single predeclared refit, $L = 2.509$, passes lensing but fails groups (Z +2.622). **No single L passes every line once the switch and the core are consistent.** The native zooms (NL1) give $L = 4.865$, with a 68% range of 3.831–5.805: not converged, and failing three cluster keys at the central value. T and L both remain **located by data**. The basin switch at a data-located $T = 10^{12.3}$ (BS1), the withdrawal of the group-sign failure (FM1), and both sealed predictions are unaffected. Section 6 gives the v1.1 additions.

Update in v1.2. IC 1613's prediction failed. The independent stellar-tracer inclination (Taibi et al. 2024, MUSE) is $36^\circ \pm 11^\circ$ and $32^\circ \pm 11^\circ$; the photometric value is $\approx 37^\circ$. That meets IN1's frozen falsifier ($i \geq 30^\circ$), so **IC 1613 is a failure** (the needed 19–27° is disfavoured at 0.5–1.4σ). **SLUGGS is partly confirmed.** Chandra Galaxy Atlas emission measures around NGC 4374 extend to 65 kpc and give $M_{\rm gas}(< 22\,{\rm kpc}) = 1.77 \times 10^9 M_\odot$, against the 1.70×10^9 extrapolation the pass needs ($f = 1.19$). NGC 5846 (atlas to 28.6 kpc) and NGC 4365 (absent) remain untested.

Abstract

This note follows PAPER63 (DOI 10.5281/zenodo.23708507), which asked where cold dark energy switches on. The programme studies a galaxy law $g_N = g^2/(g + a_0)$, with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ and $\kappa = \frac{1}{2}$ fitted. It adds a gravitating $w \approx 0$ field component called *cold dark energy*; this is not matter and not a particle.

We report frozen, reproduced calculations. Each was pre-registered and committed before its numbers, run with a mutation twin, and rerun to byte identity. Together they reduce the switch to one condition. A system carries a cored cold dark energy halo iff the collapse surface density of the *gravitational basin* it belongs to beats the galaxy floor:

$$\eta(L)\,\Sigma_{\rm coll}(M_{\rm basin}) > a_0/2\pi G.$$

The same $\max(a_0/2\pi G,\ \eta\Sigma_{\rm coll})$ sets the halo core. (v1.0 claimed the threshold mass then follows from L; that claim is withdrawn in v1.1, see the box.)

On the canonical footing this rule passes every graded calibration line:

- SPARC;
- 63 groups;
- 15 cluster lensing keys;
- five Local Group dwarf classes;
- KiDS isolated-lens lensing;
- the Milky Way.

Random settings in the explored box pass 10 times in 24.

An X-ray-bright group "wrong sign" is withdrawn as an estimator artefact. A held-out elliptical failure becomes a hot-gas prediction. IC 1613 fails on two tracers at its published inclination. v1.1 shows that both tracers are consistent with the law if $i \approx 19$–27°.

Two lensing predictions are sealed by content hash. The second is unique to the basin rule: at fixed stellar mass, a central in a massive basin should lens about half as strongly at 0.3 Mpc as an isolated one.

A candidate field equation keys condensation on the expansion of the matter flow. This is not evidence for the framework over ΛCDM.

1 The four equations

The four equations, as graded (canonical footing $a_0 = 9.3603 \times 10^{-11}\,{\rm m\,s^{-2}}$; footings never pooled):

(1) $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$.

(2) $g = \nu(g_N/a_0)\,g_N$, with $\nu(y) = \frac{1}{2} + \sqrt{\frac{1}{4} + 1/y}$.

(3) $\Sigma_{\rm core} = \max(a_0/2\pi G,\ \eta(L)\Sigma_{\rm coll})$, with $\eta = \frac{4}{3B(1)}L^{5/4}$ and $B(1) = 3\ln 2 - \pi/2$. A halo exists iff the second term wins on $M_{\rm basin}$. The concentration $c_f(M) = L\, c_{\rm nat}(M)/c_{\rm nat}(10^{14})$ follows the framework's own zoom slope. (4) The halo holds the native fair share ($\Omega_c/\Omega_b = 5.2715$, from BAO+BBN with no CMB) of the measured baryons, out to the turnaround radius.

A *basin* is the region inside the zero-velocity surface a system belongs to. For the Local Group it is measured from the radial-velocity zero crossing in the Updated Nearby Galaxy Catalog: $R_0 = 794.7 \pm 151.5\,$kpc, with $\log M_{\rm basin} = 12.392$.

2 Results (each a frozen lane; commit IDs in the repository)

Lane	Frozen verdict and key numbers
BS1 basin switch	Canonical: **resolves the Milky Way contradiction** while keeping every calibration line. The mass switch had needed $T < 10^{11.9}$ for the MW but $T > 10^{12.15}$ for lensing. The CL3 one-number variant ($L = 3.75$, $T = 10^{12.236}$) passes every canonical line.
LEE1 look-elsewhere	**BROAD**: 10/24 random (T, L) draws pass every canonical line (0.417; 68% CP [0.303, 0.539]); alt 0/24. The passing band is $L \approx 3.64$–4.17 with $T \gtrsim 10^{12.19}$. The box was not blind.
FM1 forward model	HG2 bins groups on σ, i.e. on the observed pull. Forward-modelled with the data's own scatter (0.356 dex), the base picture gives $D = -0.492$ against a measured -0.362 ($q = 0.934$): **conditioning explains the sign**. A planted $+0.4$ dex bright-group excess still gives $D < 0$, so the statistic is not diagnostic. The "wrong sign" is **withdrawn**.
SG1 SLUGGS	The IMF lever is at its dynamical bound ($M_\star/M_{\rm JAM} = 0.87$–$0.95$). Measured X-ray gas alone takes the class mean from $+0.152$ to $+0.105$ (M87 closes). Gas extrapolated beyond the measured edges passes at -0.002: **a prediction** for deeper X-ray data.
BC1 X-COP cores	The stellar profiles already carry the BCG. BCG×2 moves the $0.1\,R_{500}$ miss only from $+0.193$ to $+0.168$. The shape failure is the cluster halo's shape, not the baryons.
RT1 / EF1 IC 1613	HI rotation (LITTLE THINGS) gives $E = -0.593$ ($Z = -7.90$), matching the stellar-σ tracer (-0.535). The LG basin's external field shifts it by ≤ 0.0035 dex. **Genuine failure.**
TH1 flow basins	Basins from measured velocity zero crossings reproduce BS1's flags for all 67 Local Volume SPARC galaxies (a weak test). The MW's halo requires MW+M31 baryons $\gtrsim 1.5 \times 10^{11} M_\odot$.

3 Sealed predictions

SP1 (isolated lenses, sha256 `09cf7da6...04bf`). Above $\log M_\star \approx 11.21$, the outer $\Delta\Sigma$ of isolated centrals drops below the law's, with $S_{\rm step} = 0.47$.

SP2 (the basin step, sha256 `75b5fcdb...8b3d`). At fixed $M_\star$ below the own-mass switch, a central in a basin above T carries a cored halo truncated at R_{200}. An isolated central follows the law. The predicted ratio of their own $\Delta\Sigma$ at 0.3 Mpc, averaged over $\log M_\star$ 10.5–11.0, is $S_{\rm env} = 0.484$ (canonical; 0.442 alt). The sign is opposite to halo-model expectations for group centrals.

Both falsifiers are frozen. A KiDS-Legacy or Euclid environment split that isolates the central's own term decides them.

4 A candidate field equation (not derived, not graded)

Every selector keyed on potential depth, σ or acceleration has failed in the record, because satellites sit in shallower potentials than the Sun. A pressure fluid is excluded by the Bullet Cluster.

The basin, as measured, is where the coarse-grained expansion $\theta = \nabla \cdot v$ of the matter flow has turned negative. θ is native to Khronon/AeST-type actions through $c_2(\nabla_\mu u^\mu)^2$. A candidate order parameter χ has

$$m^2 = m_0^2\big[\Theta(\theta_u) - 1 + (a_0/2\pi G)/(\eta\Sigma_{\rm coll})\big],$$

which condenses inside turned-around regions exactly when the max(·) rule's collapse term wins. An oscillating condensate behaves as cold, collisionless matter. The step function and the smoothing scale are posited. Kill tests are listed in the repository.

5 v1.1 additions (each a frozen lane)

Lane	Frozen verdict and key numbers
CSM1 (parallel audit)	With a consistent coefficient for the switch and the core: at $L = 3.75$, $\log T \approx 11.22$ and lensing fails (+415.281). The refit $L = 2.509$ passes lensing but fails groups (+2.622). **The "T from L" claim is withdrawn.**
NL1 native L	The nine ng 64 zooms give a median $c = 7.74$ (11.63 at ng 48), so $L_{\rm nat} = 4.865$ [3.831, 5.805]. **The native level fails**: G-MID-T3, G-OUT-all and G-IN-Mall over-pull. Convergence is inconclusive: the per-seed c_f ranges over 2.4–15.1, and the estimator is binning-sensitive.
IN1 IC 1613	With LITTLE THINGS baryons and a disc estimator, both tracers match the law at one inclination, $i \approx 18.5$–$26.5°$ (Oh et al. fixed 48°; their own rings span 15–85°). **A prediction**: an independent $i \geq 30°$ keeps it a failure.
GX1 infall halo	GC3's derived infall shape on the measured-gas budget fixes the X-COP cores (+0.193 → +0.021), but fails both X-COP outskirts and lensing with the edge at $r_{\rm ta}$. With the splashback edge it passes X-COP (mean −0.035, rms 0.079) but over-masses the lensing outskirts.
SM1 samples	For the same clusters, eRASS1 $M_{\rm gas,500}$ runs 0.111 dex below X-COP's. X-COP sits at the lensing sample's 95th mass percentile. The lensing bands sit at 0.23–3.2 R_{500}, against X-COP's 0.1–0.7.
GX2 corrected gas	Raising eRASS1 gas by 0.111 dex breaks the lensing pass for every halo. **The cluster halo shape remains open.**

6 What is fitted, posited or failing

- **Fitted:** $\kappa = \frac{1}{2}$.
- **Located by data:** $L \approx 3.6$–4.2 and $T \approx 10^{12.3}$. The native test (NL1) is not converged; "T from L" is withdrawn (v1.1).
- **Posited:** the basin as the unit, i.e. the field equation above.
- **Failing:**
  - IC 1613 (independent inclination 32–36°, v1.2);
  - the cluster halo shape (X-COP against lensing);
  - the alternative a_0 footing (SPARC and the MW ultra-faints, at every setting).
- **Prediction-dependent:**
  - SLUGGS (outer hot gas; confirmed for NGC 4374 in v1.2, untested for NGC 5846 and NGC 4365).

Nothing here says the data favour the framework over ΛCDM.

AI assistance: Anthropic's Claude (analysis lanes, review, drafting). Data: SPARC; KiDS-1000; eRASS1; X-COP; SLUGGS; UNGC (Karachentsev et al. 2013); LITTLE THINGS (Oh et al. 2015); 2MRS; Kourkchi & Tully 2017. All scripts, frozen criteria and results are in the public repository.